Proving triangles similar

1

To prove: $\triangle ABC \sim \triangle XYZ$

	Statements	Reasons
1.	$\frac{AC}{XZ} = \frac{BC}{YZ}$	Given
2.	$\angle ACB \cong \angle XZY$	Given
3.	$\triangle ABC \sim \triangle XYZ$	Side-angle-side similarity (SAS $\sim$)

2

To prove: The two triangles are similar.

	Statements	Reasons
1.	$\frac{AC}{YZ} = \frac{AB}{XZ} = \frac{CB}{XY}$	Given
2.	$\triangle ABC \sim \triangle ZYX$	Side-side-side similarity (SSS $\sim$)

3

To prove: The two triangles are similar.

	Statements	Reasons
1.	$\angle IHJ \cong \angle LKJ$	Given
2.	$\angle IJH \cong \angle LJK$	Vertical angles congruence theorem
3.	$\triangle HIJ \sim \triangle KLJ$	Angle-angle similarity (AA $\sim$)

4

To prove: The two triangles are similar.

	Statements	Reasons
1.	$\frac{AC}{BC} = \frac{EC}{DC}$	Given
2.	$\angle ACE \cong \angle BCD$	Reflexive property of congruence
3.	$\triangle ACE \sim \triangle BCD$	Side-angle-side similarity (SAS $\sim$)

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To prove: The two triangles are similar.

	Statements	Reasons
1.	$\overline{AB} \parallel \overline{DE}$	Given
2.	$\angle BAC \cong \angle EDC$	Alternate interior angles theorem
3.	$\angle DCE \cong \angle ACB$	Vertical angles congruence theorem
4.	$\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)

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To prove: $\triangle ACE \sim \triangle BCD$

	Statements	Reasons
1.	$\angle CBD \cong \angle CAE$	Given
2.	$\angle BCD \cong \angle ACE$	Reflexive Property of Equality
3.	$\triangle ACE \sim \triangle BCD$	Angle-angle similarity (AA $\sim$)

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To prove: $\triangle ABC \sim \triangle DEC$

	Statements	Reasons
1.	$\frac{BC}{CE} = \frac{AC}{CD}$	Given
2.	$\angle BCA \cong \angle ECD$	Vertical angles congruence theorem
3.	$\triangle ABC \sim \triangle DEC$	Side-angle-side similarity (SAS $\sim$)

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To prove: $\triangle ABC \sim \triangle DEC$

	Statements	Reasons
1.	$\angle BAC \cong \angle EDC$	Given
2.	$\angle BCA \cong \angle ECD$	Vertical angles congruence theorem
3.	$\triangle ABC \sim \triangle DEC$	Angle-angle similarity (AA $\sim$)

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Additional questions

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a True

b False

c False

d False

e True

10

a $\triangle AYP$ is not similar to $\triangle BOX$ because $\angle A \ncong \angle B$.

b $\triangle JKL \sim \triangle JMN$ by the SAS similarity theorem.

c $\triangle UVW \sim \triangle FGH$ by the SAS similarity theorem.

d $\triangle ZYG \sim \triangle BRP$ by the SSS similarity theorem.

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a Since $\angle A \cong \angle F$ and $\angle B \cong \angle C$, $\triangle ABC \sim \triangle FCE$ by the AA similarity theorem.

b Since $\angle MVL \cong \angle UVT$ and $\frac{MV}{VT} = \frac{LV}{VU}$, $\triangle MVL \sim \triangle TVU$ by the SAS similarity theorem.

c Since $\angle U \cong \angle F$ and $\frac{UW}{HF} = \frac{UV}{FG}$, $\triangle UVW \sim \triangle FGH$ by the SAS similarity theorem.

d Since $\frac{FR}{FE} = \frac{FS}{FD} = \frac{SR}{DE}$, $\triangle FSR \sim \triangle FDE$ by the SSS similarity theorem.

12 $x = 11$

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